CHATFIELD RESEARCH

CHATFIELD
Quantitative Research

2026/27 Senior Quant Analyst Kickoff

Role • Research Cycle • Coding • GitHub • Review Process • Next Steps

Prepared for internal circulation

October 2026

ABOUT ME — JAMES DAVIES

Head of Quantitative Research



ACADEMIC BACKGROUND

- Fourth-year MSci Mathematics & Statistics, Lancaster University
- Current Master's project: *Change-Point Detection in Non-Stationary Markov Decision Processes*

RESEARCH EXPERIENCE

- **EPSRC Summer Research Internship — 2025**
  - One of three students selected for the programme
  - Research focused on changepoint detection in oil data
- **MARS Summer Research Internship — 2026**
  - Explainable image feature extraction methods
  - Worked with an industry partner as part of the research project

PREVIOUS QUANTITATIVE EXPERIENCE

- **McCormick Mentoring Scheme**
  - Mentored by a Junior Quant Developer at TP ICAP
  - Gained exposure to quantitative development in an industry setting

CHATFIELD RESEARCH

THE FOUR CHATFIELD RESEARCH SECTORS

Our Quant division will work across all four research sectors



MACRO ECONOMICS

Research into the wider economy and financial environment.

- Growth
- Inflation
- Interest rates
- Employment
- Monetary policy
- Fiscal policy



FIXED INCOME & CREDIT

Research into debt markets and the factors affecting their pricing and risk.

- Government bonds
- Corporate bonds
- Yield curves
- Interest rates
- Credit risk
- Credit spreads



COMMODITIES & ENERGY

Research into physical commodity and energy markets.

- Oil
- Gas
- Metals
- Agriculture
- Supply & demand
- Production
- Geopolitics



STUDENT ECONOMICS

Research into economic and financial issues directly affecting students.

- Cost of living
- Housing & rent
- Employment
- Student finance
- Spending

CHATFIELD RESEARCH

OUR ROLE IN CHATFIELD RESEARCH

Providing quantitative support to the research teams

FUTURE

Quantitative analysis to support the team's forward-looking research.

- Analyse relevant datasets
- Develop appropriate statistical or quantitative models
- Test relationships or assumptions
- Evaluate the evidence behind expected findings
- Provide results to inform the discussion of what happens next

CHARTS & DATA

Supporting the quantitative evidence presented in the article.

- Build charts
- Check sources
- Validate figures
- Write chart notes

OUR APPROACH

Research Question → Quantitative Approach → Analysis → Evidence for the Article

Start with the article's research question, then decide what quantitative work would genuinely help answer it.
The level and type of analysis will depend on the research question.

CHATFIELD RESEARCH

HOW I WANT THE QUANT DIVISION TO OPERATE

Rotating assignments across the four research sectors

4 SENIOR QUANT ANALYSTS

ROTATING ASSIGNMENTS
Different sectors throughout the year

4 RESEARCH SECTORS

MACRO ECONOMICS
Growth, inflation, interest rates, employment, monetary & fiscal policy.

FIXED INCOME & CREDIT
Government bonds, corporate bonds, yield curves, interest rates, credit risk.

COMMODITIES & ENERGY
Oil, gas, metals, agriculture, supply & demand, production, geopolitics.

STUDENT ECONOMICS
Cost of living, housing & rent, employment, student finance, spending.

SENIOR QUANT ANALYSTS
- Develop and implement the quantitative analysis
- Work with the assigned research team
- Take ownership of your work
- Build your skills across different sectors through rotations

HEAD OF QUANT (MY ROLE)
- Organise and manage rotations
- Support with ideas, methodology and troubleshooting
- Review the code, methodology and outputs
- Provide final technical oversight before publication

CHATFIELD RESEARCH

OUR TWO-WEEK QUANT WORKFLOW

How Quant fits into the Chatfield research cycle

WEEK 1

From article ideas to initial analysis

TUESDAY OPEN SESSION
6 PM

Hear article pitches and research plans from all teams.

→

WEDNESDAY QUANT MEETING
(≤ 1 HOUR)

Discuss potential approaches, datasets, models and feasibility.

→

DEVELOPMENT
REST OF WEEK 1

Develop the quantitative work **alongside your assigned team.**

WEEK 2

From development to publication

TUESDAY OPEN SESSION
6 PM

Hear updates from all teams and any developments.

→

WEDNESDAY QUANT MEETING
(≤ 1 HOUR)

Final checkpoint for any technical issues.

→

THURSDAY ONE-PAGER DEADLINE
Before midnight

→

FRIDAY QUANT DEADLINE

Latest deadline for completed code on GitHub.

→

SATURDAY REVIEW

I review the code, methodology and outputs.

→

SUNDAY ARTICLE DEADLINE

Article published with quantitative component reviewed.

KEY POINT

Friday is the **latest deadline.** Send your work earlier if it's ready so I have more time to review it.

CHATFIELD RESEARCH

HOW I WANT US TO WORK WITH CODE

Flexible setup, with Python as the default

PREFERRED LANGUAGE PYTHON

- Widely used in industry
- Strong ecosystem for data analysis and modelling
- Gives useful practical experience

R / RSTUDIO ALSO FINE

- If you'd prefer to work in R, I'm completely happy for you to do that.
- Focus on producing strong quantitative research, not enforcing a language.

DEVELOPMENT ENVIRONMENT

- Use whatever you're comfortable with:
  - VS Code
  - Jupyter
  - PyCharm
  - RStudio
  - Other suitable environments

INDEPENDENCE

- I trust you to manage your own development process.
- I don't need to monitor every change or approve every line of code.
- Work in the way that works best for you.

KEY PRINCIPLE

Work however you work best. Just make sure I have enough time to properly review the finished analysis.

Develop locally → **Make work accessible on GitHub** → **Review & feedback** → **Final approval**

Use your preferred environment and workflow.

Push the completed analysis when ready for review.

I review the code, methodology and outputs and provide feedback if needed.

Quantitative work approved for the article.

CHATFIELD RESEARCH

OUTPUTS AND EXPECTATIONS

Delivering high-quality, research-driven articles



RESEARCH ARTICLE

- One article per team every two weeks
- Quantitative, data-driven analysis
- Clear investment thesis and discussion



SUPPORTING ANALYSIS

- Clean, well-structured code and data
- Reproducible analysis (GitHub)
- Charts, tables and key results to support the article



GITHUB REPOSITORY

- Well-documented code
- Clear folder structure
- Reproducible analysis and outputs
- README with methodology and key findings



PRESENTATION & DISCUSSION

- Present findings at the open session and quant meeting
- Receive and incorporate feedback
- Develop communication and presentation skills

QUALITY STANDARDS



- Rigorous and well-justified analysis
- Clear and logical structure
- High-quality charts and visualisations
- Reproducible code and data
- Consideration of risks, limitations and alternative views

BENEFITS TO YOU



- Build practical quantitative research skills
- Gain experience working in a team environment
- Develop a strong portfolio of work
- Receive feedback and mentoring
- Enhance CV and future career prospects